

Exercise class 3

1. Revision of complex numbers

graphical representation, addition, multiplication, conjugation, absolute value, exponential function (e^z): Euler's formula, polar form, argument z^2 , $\sqrt{z}$ and $\log(z)$ functions

2. Prove only using real integrals and the Euler form of the exponential function that for any $z \in \mathbb{C}$, $z \neq 0$.

$$\int e^{zx} dx = \frac{1}{z} e^{zx}$$

3. Calculate the complex Fourier transforms of the functions below.

a. $f(x) = \begin{cases} 1 & \text{if } b < x < c \\ 0 & \text{otherwise} \end{cases}$

b. $f(x) = \Theta(x)e^{-ax} = \begin{cases} e^{-ax} & \text{if } 0 < x \\ 0 & \text{if } x < 0 \end{cases}$

c. $f(x) = e^{-a|x|}$

d. $f(x) = \frac{1}{x^2 + a^2}$

e. $f(x) = \delta(x - x_0)$

f. $f(x) = e^{i\omega_0 x}$

g. $f(x) = e^{-a x^2}$

h. $f(x) = e^{-a(x-b)^2}$

i. $f(x) = e^{-a x^2 + i b x}$